

# DOKUMENTACJA GAIA

## 1. Opis i Wymagania

Proponowane rozwiązanie zakłada wykorzystanie systemu agentowego, który ma na celu poprawę dopasowania lokali usługowych do potrzeb lokalnych społeczności, wspierając realizację koncepcji „miasta 15-minutowego”. System ten składa się z autonomicznych agentów, którzy reprezentują różne podmioty: właścicieli lokali do wynajęcia, potencjalnych najemców, lokalnych mieszkańców oraz jednostki agregujące dane o lokalach w danej lokalizacji. Dzięki temu możliwe jest optymalne rozmieszczenie usług, co przyczynia się do wzmocnienia lokalnych społeczności oraz zwiększenia efektywności rynku wynajmu lokali.

- System umożliwia parowanie lokali z biznesami poprzez aukcje, parowanie na podstawie (z zachowaniem koncepcji miasta 15-minutowego). Istotne czynniki parowania stanowią:• cena,• lokalizacja,• rozmiar,• typ działalności.
- System umożliwia po sparowaniu akceptację lub odrzucenie dobranej pary przez biznes lub lokal
- System umożliwia dodanie lokalu do giełdy
- System umożliwia usuwanie lokalu z giełdy.
- System umożliwia złożenie zapotrzebowania na lokal przez biznes.
- System umożliwia podanie górnej i dolnej granicy dla kryteriów parowania.
- System powinien umożliwiać złożenie zapotrzebowania na usługę przez mieszkańca
- System powinien umożliwiać wycofanie oferty przez lokal
- System mógłby umożliwiać wycofanie zapotrzebowania na usługę przez mieszkańca.

## 2. Model Ról

### Role: PropertyOwner

|                  |                                                                                                                                                                        |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description      | Entity representing a property owner who can list properties for rent or sale in the marketplace, managing real estate assets and their availability for business use. |
| Responsibilities | <b>Liveness:</b> (ListProperty . RemoveProperty . WithdrawPropertyOffer)^ω<br><b>Safety:</b> UniqueID(PropertyID) != Null<br>PropertyID in PropertyList                |
| Permissions      | READ: PropertyID<br>WRITE: PropertyID<br>CONSUME: ListingConfirmation, RemovalConfirmation, WithdrawalConfirmation<br>PRODUCE: PropertyID                              |
| Activities       |                                                                                                                                                                        |
| Protocols        | ListProperty, RemoveProperty, WithdrawPropertyOffer                                                                                                                    |

### Role: BusinessEntity

|                  |                                                                                                                                                                                                                                                                                                                                                                              |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description      | Entity representing a business that seeks suitable premises for its operations, including requirements related to location, size, and activity type, and actively engages in the matching process with property owners.                                                                                                                                                      |
| Responsibilities | <b>Liveness:</b> (MatchBusinessWithProperty . RespondToMatch . SubmitBusinessDemand . SetMatchingBoundaries)^ω<br><b>Safety:</b> UniqueID(BusinessID) != Null<br>BusinessID in BusinessList<br>MatchCriteria.Valid<br>DemandCriteria.Valid                                                                                                                                   |
| Permissions      | READ: BusinessID, PropertyID, MatchCriteria, DemandCriteria, BoundaryCriteria<br>WRITE: BusinessID, PropertyID, MatchCriteria, DemandCriteria, BoundaryCriteria<br>CONSUME: MatchAcceptance, MatchRejection, MatchConfirmation, MatchRejection, DemandConfirmation, BoundaryConfirmation<br>PRODUCE: BusinessID, PropertyID, MatchCriteria, DemandCriteria, BoundaryCriteria |
| Activities       |                                                                                                                                                                                                                                                                                                                                                                              |
| Protocols        | MatchBusinessWithProperty, RespondToMatch, SubmitBusinessDemand, SetMatchingBoundaries                                                                                                                                                                                                                                                                                       |

### Role: Resident

|                  |                                                                                                                                                                  |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description      | Entity representing a local resident who may express demand for services or amenities in their neighborhood, contributing to community-driven service needs.     |
| Responsibilities | <b>Liveness:</b> (SubmitResidentDemand . WithdrawResidentDemand)^ω<br><b>Safety:</b> UniqueID(ResidentID) != Null<br>ResidentID in ResidentList                  |
| Permissions      | READ: ResidentID, ServiceDemand<br>WRITE: ResidentID, ServiceDemand<br>CONSUME: DemandConfirmation, WithdrawalConfirmation<br>PRODUCE: ResidentID, ServiceDemand |
| Activities       |                                                                                                                                                                  |
| Protocols        | SubmitResidentDemand, WithdrawResidentDemand                                                                                                                     |

## Role: Marketplace

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description      | Entity responsible for managing the marketplace, including listing properties, matching businesses with properties, and maintaining data integrity across all transactions.                                                                                                                                                                                                                                                                                                                                                     |
| Responsibilities | <b>Liveness:</b> (ManagePropertyListings    ProcessBusinessDemands    ExecuteMatchingAlgorithms    ValidateDataIntegrity)^ω<br><b>Safety:</b> UniqueID(PropertyID) != Null<br>UniqueID(BusinessID) != Null<br>UniqueID(ResidentID) != Null<br>PropertyID in PropertyList<br>BusinessID in BusinessList<br>ResidentID in ResidentList<br>MatchCriteria.Valid<br>DemandCriteria.Valid<br>BoundaryCriteria.Valid                                                                                                                   |
| Permissions      | READ: PropertyID, BusinessID, ResidentID, MatchCriteria, DemandCriteria, BoundaryCriteria, ServiceDemand<br>WRITE: PropertyID, BusinessID, ResidentID, MatchCriteria, DemandCriteria, BoundaryCriteria, ServiceDemand<br>CONSUME: PropertyID, BusinessID, ResidentID, MatchCriteria, DemandCriteria, BoundaryCriteria, ServiceDemand<br>PRODUCE: ListingConfirmation, RemovalConfirmation, MatchAcceptance, MatchRejection, MatchConfirmation, MatchRejection, DemandConfirmation, BoundaryConfirmation, WithdrawalConfirmation |
| Activities       | ManagePropertyListings, ProcessBusinessDemands, ExecuteMatchingAlgorithms, ValidateDataIntegrity                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Protocols        | ListProperty, RemoveProperty, MatchBusinessWithProperty, RespondToMatch, SubmitBusinessDemand, SetMatchingBoundaries, SubmitResidentDemand, WithdrawPropertyOffer, WithdrawResidentDemand                                                                                                                                                                                                                                                                                                                                       |

### 3. Model Interakcji

| Global                                | Name                      | Initiator      | Responder      | Inputs                                | Outputs                           | Description                                                                                                                                           |
|---------------------------------------|---------------------------|----------------|----------------|---------------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Property on the marketplace           | ListProperty              | PropertyOwner  | Marketplace    | PropertyID                            | ListingConfirmation               | Initiated by PropertyOwner to add a property to the marketplace.                                                                                      |
| Remove property from marketplace      | RemoveProperty            | PropertyOwner  | Marketplace    | PropertyID                            | RemovalConfirmation               | Initiated by PropertyOwner to remove a property from the marketplace.                                                                                 |
| Business match proposal               | MatchBusinessWithProperty | Marketplace    | BusinessEntity | BusinessID, PropertyID, MatchCriteria | MatchAcceptance, MatchRejection   | Initiated by Marketplace to propose a match between a business and a property based on criteria such as price, location, size, and activity type.     |
| Respond to match proposal             | RespondToMatchProposal    | BusinessEntity | Marketplace    | BusinessID, PropertyID, Response      | MatchConfirmation, MatchRejection | Initiated by BusinessEntity to respond to a match proposal from the Marketplace to accept or reject the proposed match.                               |
| Submit demand for service             | SubmitBusinessDemand      | BusinessEntity | Marketplace    | BusinessID, DemandCriteria            | DemandConfirmation                | Initiated by BusinessEntity to submit a demand to the Marketplace to submit a demand for a service at a specific property based on specific criteria. |
| Set matching criteria                 | SetMatchingBoundaries     | Marketplace    | BusinessEntity | BusinessID, BoundaryCriteria          | BoundaryConfirmation              | Initiated by Marketplace to set upper and lower boundaries for matching criteria with BusinessEntity.                                                 |
| Submit demand for service at property | SubmitResidentDemand      | Resident       | Marketplace    | ResidentID, ServiceDemand             | DemandConfirmation                | Initiated by Resident to submit a demand for a service at a specific property.                                                                        |
| Withdraw property from marketplace    | WithdrawPropertyOffer     | PropertyOwner  | Marketplace    | PropertyID                            | WithdrawalConfirmation            | Initiated by PropertyOwner to withdraw a property offer from the marketplace.                                                                         |
| Withdraw service demand               | WithdrawResidentDemand    | Resident       | Marketplace    | ResidentID, ServiceDemandID           | WithdrawalConfirmation            | Initiated by Resident to withdraw a service demand from the marketplace.                                                                              |

## 4. Model Usage

| Service                   | Agent               | In/Out                                                                                          | Pre/Post Condition                                                                                                                                                                                                         | Derived From              |
|---------------------------|---------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| ListProperty              | PropertyOwnerAgent  | <b>IN:</b> PropertyID<br><b>OUT:</b> ListingConfirmation                                        | <b>PRE:</b> PropertyID is valid and not already listed<br><b>POST:</b> PropertyID is added to PropertyList                                                                                                                 | ListProperty              |
| RemoveProperty            | PropertyOwnerAgent  | <b>IN:</b> PropertyID<br><b>OUT:</b> RemovalConfirmation                                        | <b>PRE:</b> PropertyID exists in PropertyList<br><b>POST:</b> PropertyID is removed from PropertyList                                                                                                                      | RemoveProperty            |
| WithdrawPropertyOffer     | PropertyOwnerAgent  | <b>IN:</b> PropertyID<br><b>OUT:</b> WithdrawalConfirmation                                     | <b>PRE:</b> PropertyID exists in PropertyList<br><b>POST:</b> PropertyID is removed from PropertyList and no longer available for matching                                                                                 | WithdrawPropertyOffer     |
| MatchBusinessWithProperty | MarketplaceAgent    | <b>IN:</b> BusinessID, PropertyID, MatchCriteria<br><b>OUT:</b> MatchAcceptance, MatchRejection | <b>PRE:</b> BusinessID is valid and exists in BusinessList, PropertyID is valid and exists in PropertyList, MatchCriteria are valid and within defined boundaries<br><b>POST:</b> Match proposal is sent to BusinessEntity | MatchBusinessWithProperty |
| RespondToMatch            | BusinessEntityAgent | <b>IN:</b> BusinessID, PropertyID, Response<br><b>OUT:</b> MatchConfirmation, MatchRejection    | <b>PRE:</b> Match proposal exists for BusinessID and PropertyID<br><b>POST:</b> Match result is confirmed or rejected                                                                                                      | RespondToMatch            |
| SubmitBusinessDemand      | BusinessEntityAgent | <b>IN:</b> BusinessID, DemandCriteria<br><b>OUT:</b> DemandConfirmation                         | <b>PRE:</b> BusinessID is valid and DemandCriteria are valid<br><b>POST:</b> DemandCriteria are stored in BusinessList                                                                                                     | SubmitBusinessDemand      |
| SetMatchingBoundaries     | MarketplaceAgent    | <b>IN:</b> BusinessID, BoundaryCriteria<br><b>OUT:</b> BoundaryConfirmation                     | <b>PRE:</b> BusinessID is valid and BoundaryCriteria are valid<br><b>POST:</b> BoundaryCriteria are stored for BusinessID and used in matching algorithms                                                                  | SetMatchingBoundaries     |
| SubmitResidentDemand      | ResidentAgent       | <b>IN:</b> ResidentID, ServiceDemand<br><b>OUT:</b> DemandConfirmation                          | <b>PRE:</b> ResidentID is valid and ServiceDemand is valid<br><b>POST:</b> ServiceDemand is stored in ResidentList                                                                                                         | SubmitResidentDemand      |
| WithdrawResidentDemand    | ResidentAgent       | <b>IN:</b> ResidentID, ServiceDemandID<br><b>OUT:</b> WithdrawalConfirmation                    | <b>PRE:</b> ResidentID is valid and ServiceDemandID exists in ResidentList<br><b>POST:</b> ServiceDemandID is removed from ResidentList                                                                                    | WithdrawResidentDemand    |
| ManagePropertyListings    | MarketplaceAgent    | <b>IN:</b><br><b>OUT:</b>                                                                       | <b>PRE:</b> Marketplace is operational<br><b>POST:</b> Property listings are managed correctly                                                                                                                             | ManagePropertyListings    |
| ProcessBusinessDemands    | MarketplaceAgent    | <b>IN:</b><br><b>OUT:</b>                                                                       | <b>PRE:</b> Marketplace is operational<br><b>POST:</b> Business demands are processed correctly                                                                                                                            | ProcessBusinessDemands    |
| ExecuteMatchingAlgorithms | MarketplaceAgent    | <b>IN:</b> BusinessID, PropertyID, MatchCriteria<br><b>OUT:</b> MatchAcceptance, MatchRejection | <b>PRE:</b> BusinessID and PropertyID are valid and exist in respective lists, MatchCriteria are valid and within defined boundaries<br><b>POST:</b> Matching algorithms are executed and results are returned             | ExecuteMatchingAlgorithms |

## 5. Model Znajomości

